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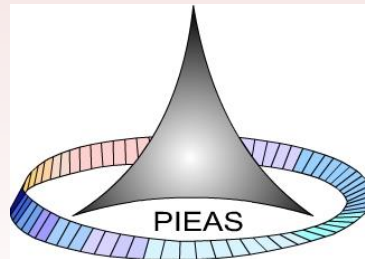
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Application of Information and Communication Technologies CIS-134T

Artificial Intelligence Viewports with Machine Learning (Chapter 15)



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Course Outline



Number Systems, Binary numbers, Boolean logic, History computer system, basic machine organization, Von Neumann Architecture, Algorithm definition, design, and implementation, Programming paradigms and languages, Graphical programming, Overview of Software Engineering and Information Communication Technology, Operating system, Compiler, DBMS, Computer networks and internet, WWW, web mail applications, Computer graphics, **Artificial Intelligence**, Computer Vision, Cybersecurity, Viruses and Anti-Viruses, Cloud Computing, **Use of office productivity tools, such as word processors**, spreadsheets, presentation applications, etc., Social, Ethical, Professional and Legal Issues. **Related Lab and practical work against the mentioned topics.**



Chapter No. 15

Artificial Intelligence



Goals



- Artificial Intelligence (AI)
- Machine Learning (ML)
- Modular Examples (BTsClSeg, BrCan, PhySec)



What is AI?

- Artificial Intelligence (AI) is the simulation of human intelligence in machines that can perform tasks like learning, reasoning, and problem-solving
- Examples: Siri (Apple), Alexa (Amazon), Self-driving cars
- *Alexa can play your favorite song, read the latest headlines, dim the lights in your living room, and more.*





What is AI?



CIS-134T



History of AI

- 1943 McCulloch & Pitts: Boolean circuit model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
- 1965 Robinson's complete algorithm for logical reasoning
- 1966—73 AI discovers computational complexity
Neural network research almost disappears
- 1969—79 Early development of knowledge-based systems
- 1980-- LISP based machines developed and marketed
- 1986-- Neural networks return to popularity
- 1987-- AI becomes a science
- 1995-- The emergence of intelligent agents



Introduction



- What is intelligence?
 - Humans?
 - Behave intelligently as a human
 - Behave in the best possible manner
 - Thinking? Related to *reasoning*
 - Acting? Related to *behavior*
- What is artificial intelligence?
 - is concerned with the design of intelligence in an artificial device (McCarthy, 1956)
- Why the word artificial?
- Compare humans with animals with respect to intelligence?



Introduction



- What is intelligence?
 - the ability to learn facts and skills and apply them
- What is artificial intelligence?
 - is concerned with the design of intelligence in an artificial device (McCarthy, 1956)
 - Science & engineering of making intelligent machines, especially intelligent computer programs
 - Branch of computer science that is concerned with the automation of intelligent behavior
 - Automation? replacement of human workers by technology



Example



- What actually are we doing?
 - “ We are putting some constraints, or finding features corresponding to this photograph being searched”
- *In computer language*: making a tree corresponding to this database file of all criminals so that identification can be carried out faster
- There can be many constraints: such as color of hair, eyes, skin, etc.
- CAN YOU IDENTIFY *AGENT* HERE IN THIS EXAMPLE?



Example



- Example related to identify a criminal.
- CAN YOU IDENTIFY *AGENT* HERE IN THIS EXAMPLE?



What is AI?

- *Goal* : Identify a criminal out of 0.1 million files
- Each criminal has the data like *name, height, sex, age, gender, address*, etc.
- Just start matching the criminal to one of the entity in the data file of criminals
- It is quite exhaustive for humans!
- It has been observed that eyes start losing sensitivity to differentiate photographs after about how many photos in the file?
- 50 to 60
- What about a machine to do this job?



What is AI?

- But this machine has many components.
- Firstly, the data file must be computerized or in the form of a computer file with all records
- For any new photographs, the machine has to do the matching
- How?
- Devise a strategy that these photographs are not to be seen or matched
- Luckily there exists a method, e.g., the height is $< 5'7"$ (no need to see the photographs of all the criminals)
- Also, if the photograph being searched is a male, then no need to search/ compare with the photographs of females



What is AI?

- Another example is a classification of pixels in satellite images.
- What is a *satellite image*?
- It is a photograph of the Earth acquired from the satellite with a resolution of pixel size 36 m^2
- *One of the main problems of satellite images:* segregation of pixels, e.g., the size of the image is 512 (rows) X 512 (cols)
- Each pixel belongs to a land cover type, e.g., a pixel may correspond to a water region or a hilly region; maybe there is a building at the location of the pixel



What is AI?

- There are many land cover types, such as rice fields, open spaces (barren land), concrete structures, forests, sand (desert) area, sea area, etc.
- *Question*: How many land cover types may be possessed by the region under consideration?
- Why I need to do this?



What is AI?



- It has many uses (suggest some)
 - Complete geology of a location is needed in many places, e.g., metro services in a city
 - Civil uses: How much forest area is being depleted each year? It has long term impact
 - Defense applications: Movement of enemy troops in the vicinity of border areas, amount of new constructions taking place (*every state would be interested in knowing this*)
 - How much area is devastated in the floods? One may prove by using photographs before and after floods
 - AI techniques can be used to find out the amount of area being devastated by natural calamities like floods, draught, earthquakes, etc.
- Humans cannot do this!



Human Intelligence

- Differentiation between text marker and snake, and see the purpose of board marker and what a snake can do? Make a decision whether walk over it or not
- What a robot will do in such a situation? (as we have no knowledge about environment there)
- Robot while moving should know to differentiate a camera, a bulb, a can, a computer, a motorbike, etc.



Human Intelligence

- Different locations of switch boards in rooms, located on left or right
- A robot should know what everything on earth is like we understand.
- Take a look at table, and carpet. We shall be more careful while walking on a carpet than on a hard surface
- If any of the things is harmful for the robot itself?



Human Intelligence

- How does a robot know what to do on a mars surface?
- “Problem of identifying each and every thing we see”
- The objective is to make computer do everything which human beings can logically think of
- In medical image processing, people are trying to see what is happening inside the body
- Several types of images, scans, EEG, ECG, ultrasound machines are used for medical images



Human Beings Learn

- Ask a person to find an address (a house in city)
- He would learn it. He would not have read about it
- THIS IS AN EXAMPLE OF LEARNING
- By using senses, brain and learning process we are making a machine which is replacing human beings
- Human beings have another characteristic: human beings are emotional
- When we are emotional, we are not able to think logically
- If we make a machine, that does not have emotion characteristic, then that machine is much better than human beings



Viewpoints of AI

Views of AI fall into four categories:

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

Modern approach advocates "*acting rationally*" (logical reasoning)



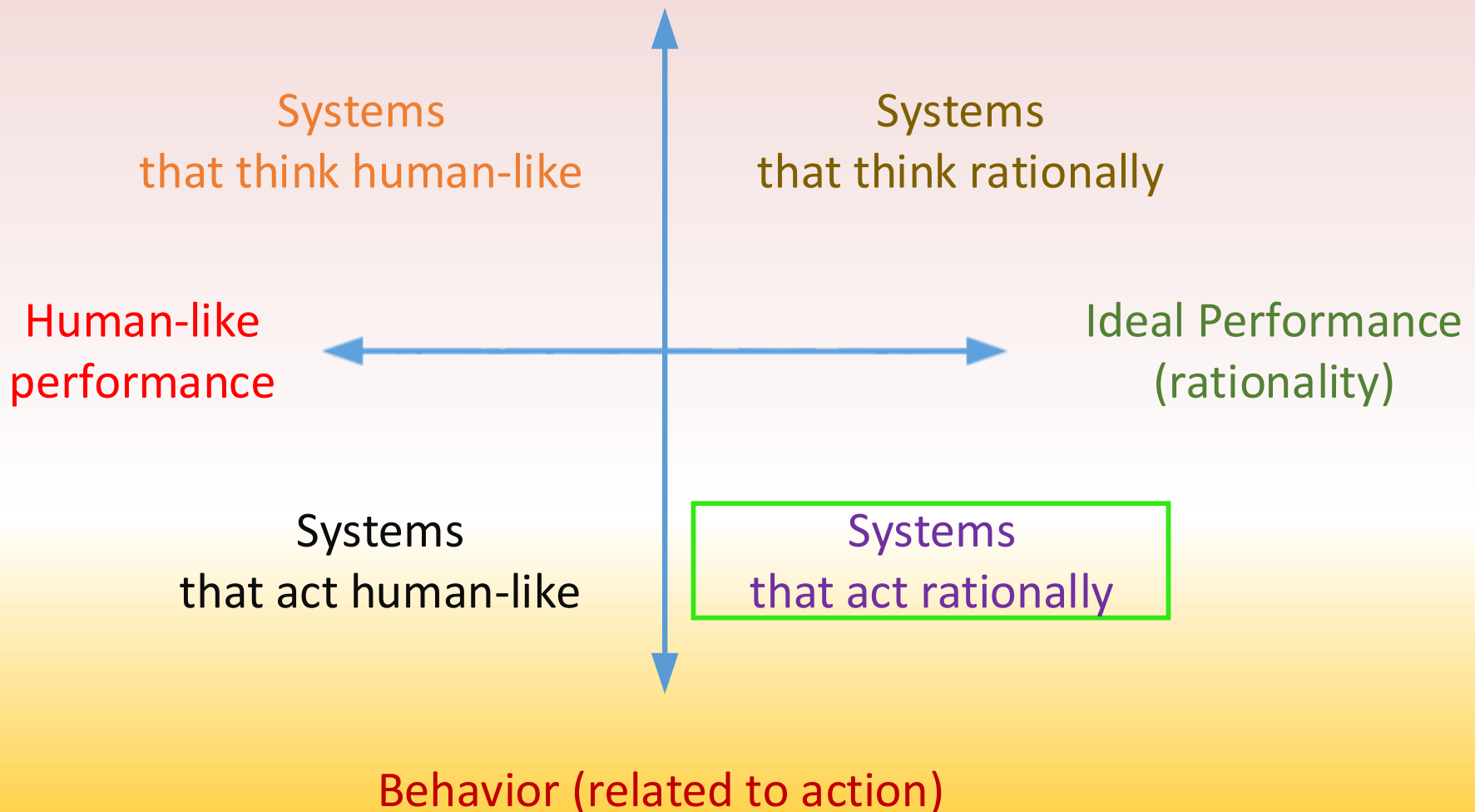
Viewpoints of AI

- AI – concerned with intelligence, where humans (i.e., homo sapiens) are considered a supreme being
 - *Controversy*: can we create intelligent machines?
- Definitions fall in different categories according to the viewpoint of the definer
 - **Acting (apparent behavior) or Thinking (thought processes & reasoning)**
 - *Measure*: Human or Rationality (reasonable & sensible)
 - ❖ Rationality: an ideal concept of intelligence. A system is Rational if it does the “right thing,” given what it knows



Some General Trends to AI

Thought / Reasoning





Some Definitions of AI

Systems that think like humans	Systems that think rationally
"The exciting new effort to make computers think . . . <i>machines with minds</i>, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Chamiak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Systems that act like humans	Systems that act rationally
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole <i>et al.</i>, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)



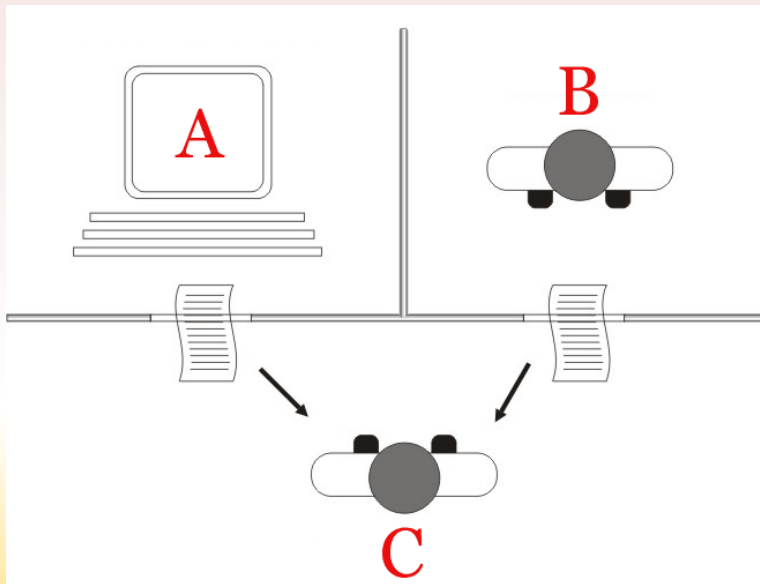
What is AI?

- The ones on top are concerned with *thought processes* and *reasoning*, whereas the ones on the bottom address *behavior* (*actions*)
- The definitions on the left measure success in terms of fidelity to *human* performance, whereas the ones on the right measure against an *ideal* concept of intelligence, which we will call rationality



Acting Humanly: Turing Test

- Alan Turing (1950) "Computing machinery and intelligence"
- Turing test: the system which passed the Turing test would be a system that behaves like a human or thinks like a human
- "Can machines think?" → "Can machines behave intelligently?"



A : a computer being interrogated

B : a human

C : interrogator

- Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes



The Turing test “Result”

If the interrogator cannot reliably distinguish the human from the computer then the computer does possess (artificial) intelligence



Acting Humanly: Turing Test

- The *computer passes the test* if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or not
- The computer would need to possess the capabilities by suggesting major components of AI:
 - 1) **knowledge** representation to store what it knows or hears
 - 2) automated **reasoning** to use the stored information to answer questions and to draw new conclusions;
 - 3) **natural language processing** to enable it to communicate successfully in English
 - 4) **machine learning** to adapt to new circumstances and to detect and extrapolate patterns
 - 5) **computer vision** to perceive objects (*Total Turing Test* only)
 - 6) **robotics** to manipulate objects and move about (*Total Turing Test* only)



Problem with Turing Test

It is not **reproducible**, **constructive**, or
amenable to mathematical analysis



Acting Rationally: Rational Agent

- *Rational behavior*: doing the right thing
- The right thing: that which is expected to maximize goal achievement, **given the available information**
- Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action
- Aristotle (Nicomachean Ethics):

Every art and every inquiry, and similarly every action and pursuit, is thought to aim at some good



What We Call Easy?

- It has been easier to mechanize many of the high-level tasks we usually associate with “intelligence” in people
 - Symbolic integration
 - Playing chess
 - Medical diagnosis, such as tumor types (malignant/benign)
 - Proving mathematical theorems



What We Call Hard?



- It has been very hard to mechanize tasks that lots of animals can do
 - Walking around without running into things
 - Interpreting complex sensory information
 - Catching prey and avoiding predators
 - Modelling the internal states of other animals from their behavior



Approaches to AI

- *Applied AI*: aims to produce commercially viable “smart” systems – such as, for example, a security system that is able to recognize the faces of people who are permitted to enter a particular building. Applied AI has already enjoyed considerable success.
- *Cognitive AI*: computers are used to test theories about how the human mind works – for example, theories about how we recognized faces and other objects, or about how we solve an abstract problem



What AI Systems Can Do?

- Spoken language: 1000 words continuous speech
- Planning and scheduling: Hubble Telescope experiments
- Learning: text categorization into ~1000 topics
- Games: Grand Master level in Chess, checkers, cricket
- Detection of tumor in a diseased subject
- E-medicine and E-radiology via DICOM and PACS
- Reconstruction of limbs/ body parts by 3D printing
- Simulating any situation without actually performing it



What AI Systems Can't Do Yet?



- Understand natural language (read and understand articles in a newspaper)
- Web surfing
- Understand the language of animals
- Perception of animal viewing abilities
- Interpret an arbitrary visual scene
- Learn a natural language
- Construct plans in dynamic real-time domains
- Exhibit true autonomy and intelligence



AI Prehistory

- Philosophy Logic, methods of reasoning, mind as physical system foundations of learning, language, rationality
- Mathematics Formal representation and proof algorithms, computation, (un)decidability, (in)tractability, probability
- Economics utility, decision theory
- Neuroscience physical substrate for mental activity
- Psychology phenomena of perception and motor control, experimental techniques
- Computer engineering building fast computers
- Control theory design systems that maximize an objective function over time
- Linguistics knowledge representation, grammar



State of the art

- **Deep Blue defeated** the reigning world chess champion Garry Kasparov in 1997
- Proved a mathematical conjecture (**Robbins conjecture**) unsolved for decades
- **No hands** across America (driving autonomously 98% of the time from Pittsburgh to San Diego)
- During the 1991 Gulf War, US forces deployed an AI **logistics planning and scheduling program** that involved up to 50,000 vehicles, cargo, and people
- NASA's on-board autonomous planning program controlled the **scheduling of operations** for a spacecraft
- **PROVERB** solves crossword puzzles better than most humans (Duke Uni)



Association for the Advancement of Artificial Intelligence (AAAI)



- Founded in 1979, the Association for the Advancement of Artificial Intelligence (AAAI) (formerly the **American Association for Artificial Intelligence**) is a nonprofit scientific society devoted to **advancing the scientific understanding of the mechanisms underlying thought and intelligent behavior and their embodiment in machines**
- <http://www.aaai.org/>
- Information about AI from the News, Publications, and Conferences
- <https://aitopics.org/search>



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Information about AI from the News, Publications, and Conferences

Automatic Classification – Tagging and Summarization – Customizable Filtering and Analysis

If you are looking for an answer to the question *What is Artificial Intelligence?* and you only have a minute, then here's the definition the [Association for the Advancement of Artificial Intelligence](#) offers on its home page: "the scientific understanding of the mechanisms underlying thought and intelligent behavior and their embodiment in machines."

However, if you are fortunate enough to have more than a minute, then please get ready to embark upon an exciting journey exploring AI (but beware, it could last a lifetime) ...

Modeling Uncertainty Helps MIT's Drone Zip Around Obstacles

 **IEEE Spectrum Robotics Channel** 30 minutes ago 

It's not too hard to make a drone that can fly very fast, and it's not too hard to make a drone that can avoid obstacles. Making a drone that can do both at once is much more difficult, but it's necessary in order for them to be real-world useful. At MIT CSAIL, Pete Florence (in Russ Tedrake's lab)...

 [artificial intelligence, drone, obstacle, \(14 more...\)](#)

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Google releases Cloud TPU beta, GPU support for Kubernetes



Branches of AI

- Machine Learning
- Speech synthesis and recognition
- Natural language processing and understanding
- Problem solving
- Game playing
- Expert systems
- Machine vision



Assignment

1. Define intelligence
2. What are different approaches in defining AI?
3. Let us suppose you design a Turing machine. What capabilities would you like to incorporate in it?
4. Design ten questions to pose to a human/ machine that is taking the Turing test.
5. Will building an AI computer automatically shed light on the nature of natural intelligence?
6. List 5 tasks that you will like a computer to be able to do within the next 5 years.
7. List 5 tasks that computers are unlikely to be able to do in the next 10 years.
8. Define (i) an agent, (ii) intelligent agent, (iii) rational agent, (iv) bounded rationality
9. Discuss different types of environments.
10. Explain different agent architectures.

Machine Learning



What is Machine Learning?

- Introduction
- Key Concepts and Terminology
- Components of a Learning System
- Training and Testing Phases
- Overfitting and Underfitting
- Features Selection
- ML Goals
- Types of Machine Learning
- Challenges and Future Trends



Machine Learning



Learning is the ability to improve one's behavior based on experiences

ML is related with making computer systems that automatically improve with experience



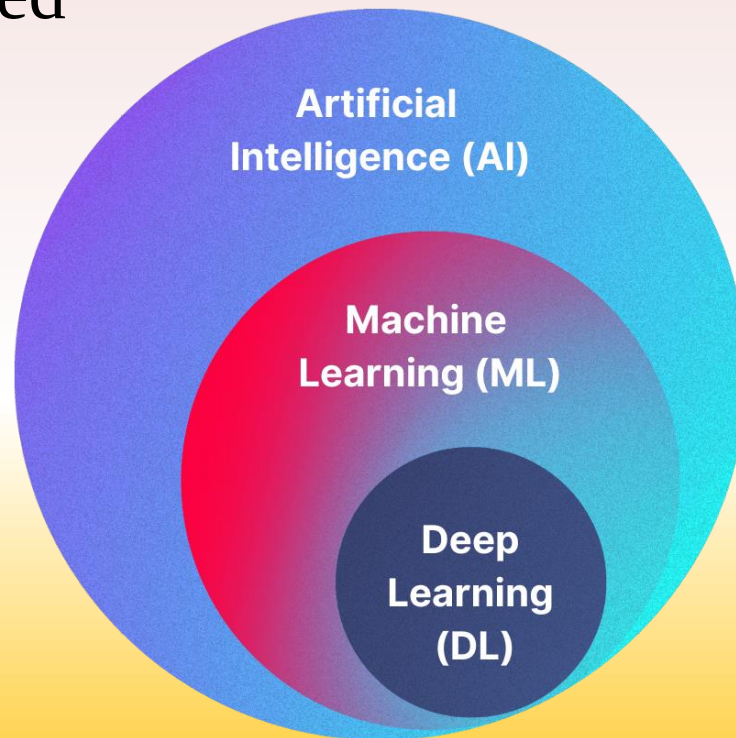
What is ML?

- Machine learning is programming computers to optimize a performance criterion using example data or experience
- We have a model defined up to some parameters
- Learning is the execution of a computer program to optimize the parameters of the model using the training data or experience.
- The model may be predictive to make predictions in the future, or descriptive to gain knowledge from data, or both.



What is ML?

- Machine learning is the key enabler of AI
- It is the field of study that gives computers the capability to learn without being explicitly programmed





Applications

- Self driving cars
- Automatic speech recognition system
- Law, science, business industry for solving important tasks arising from time to time
 - Financial, commercial, medical and scientific community applications
- Vastly improved understanding of the human genome



ML – a question

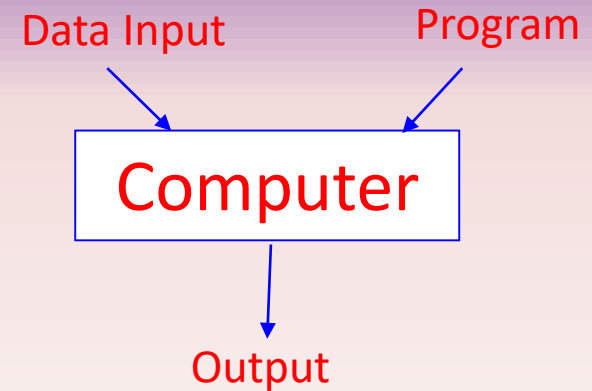


How a machine learning solution differs from
a programmatic solution?

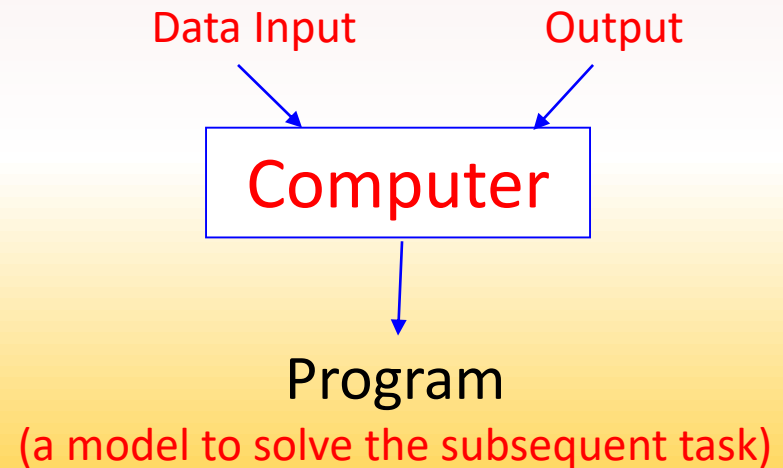


ML – a question

Algorithm to solve a problem



Machine Learning





Components of a Learning Algorithm

- **E**, experience, input or the training data
- **T**, prediction, classifying/ acting in an environment
- **P**, measure of performance
- *Example:* one may want to improve the efficiency of problem solving or the objective may be to increase the prediction accuracy

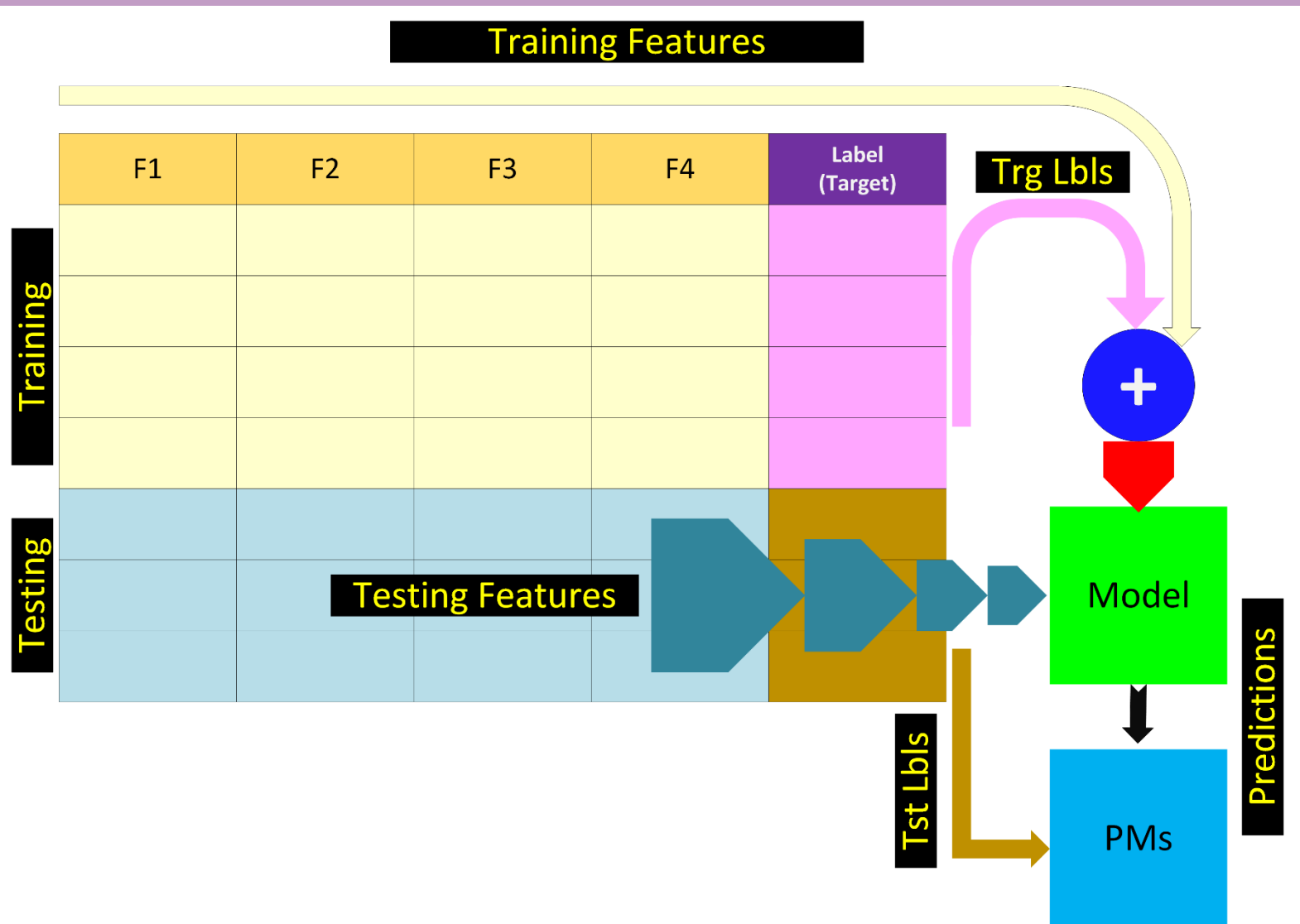


ML – fundamentals

- Formally, ML may be thought of as follows
 - A computer program is said to learn from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, improves with experience



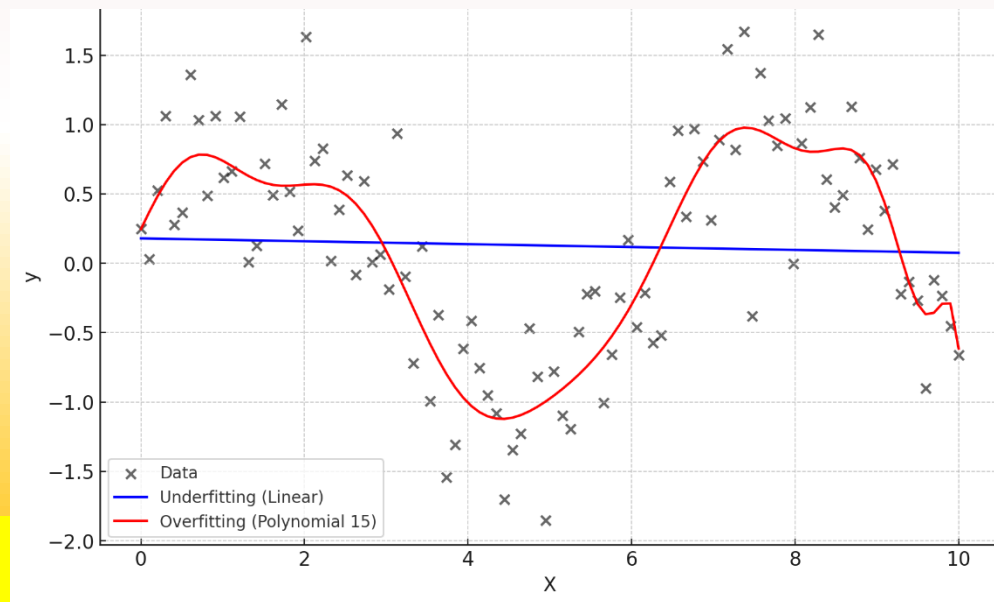
Training and Testing Phases





Underfitting and Overfitting

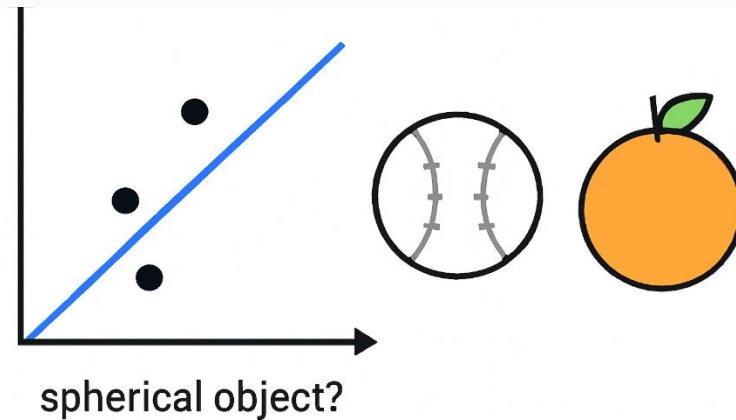
- The blue line represents underfitting, where a simple linear model fails to capture the complexity of the data
- The red line represents overfitting, where a high-degree polynomial model fits the noise in the data too closely, creating a curve that is too complex for the underlying trend
- In real-world scenarios, we aim for a model that captures the underlying patterns without being overly complex or too simple (best fit)





Underfitting

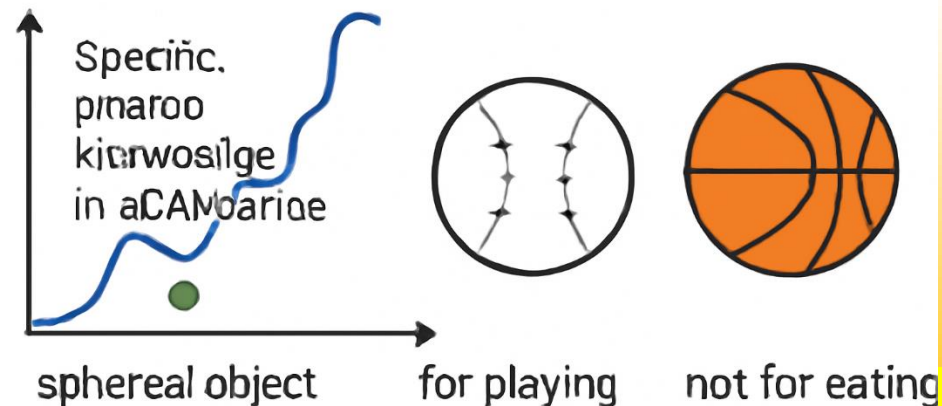
- Less knowledge, as limited points, follows the line
- A lesser number of measurements (features)
- Goal: find the ball
- Single feature used: the object is spherical
- A test is carried out on an orange: it is spherical but not a ball, and cannot be used for playing





Overfitting

- It tries to cover every point of the data
- Specific, precise knowledge in abundance
- Four features used: Spherical object, for playing, not for eating, and radius $< 5\text{cm}$
- Test case: a basketball
- The "Best Fit" model captures the general trend of the data without being too simple (underfitting) or too complex (overfitting).





Feature Selection

- Feature selection reduces the **dimensionality of data** by selecting only a subset of measured features (predictor variables) to create a model.
- Selection criteria usually involve the **minimization** of a specific measure of **predictive error for models fit** to different subsets.
- Algorithms search for a subset of predictors that optimally model measured responses, **subject to constraints** such as **required or excluded features** and **the size of the subset**.



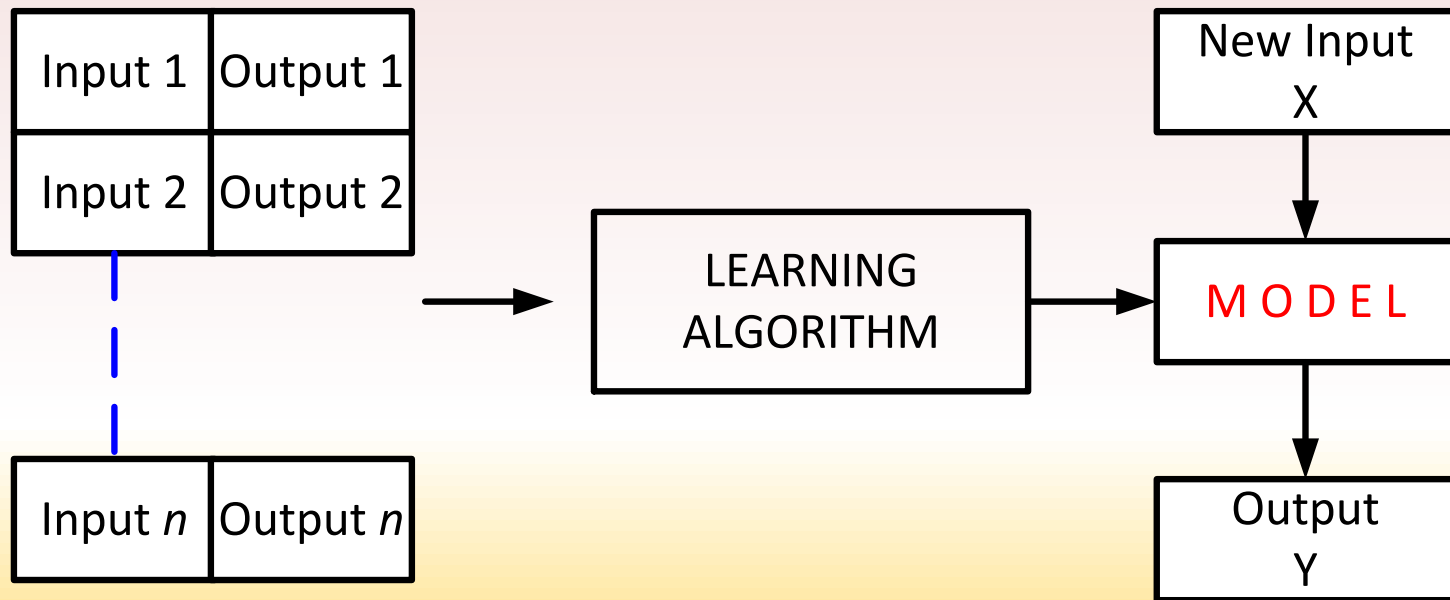
ML - Goals

- Generally aim to improve the performance of systems through learning from data.
- To predict outcomes or future events based on historical data (predicting house prices, stock prices) SL
- To classify data into predefined classes or labels (Email spam detection, sentiment analysis, tumor classification) SL
- To group similar data points (forming clusters) based on features, without prior knowledge of labels (Market segmentation, grouping customers with similar buying behavior) USL
- To identify rare or unusual patterns (anomaly detection) in data that deviate significantly from the norm (Fraud detection, identifying network intrusions), USL
- To reduce the number of features in data while preserving its essential structure using dimensional reduction techniques (pca, lda, ica) USL

Types of ML

Supervised Learning: scheme

DATA

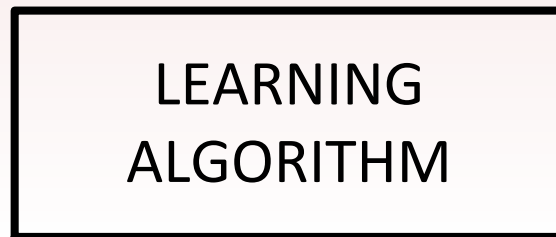
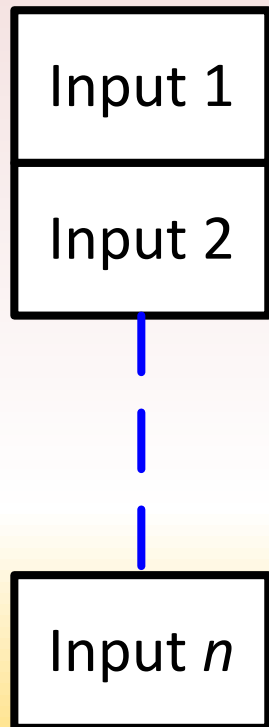


Supervised Learning: scheme

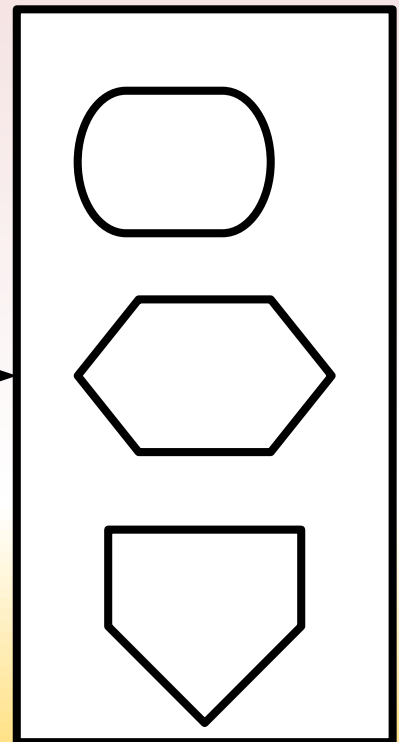
- For every data instance we have input and output
- ML system builds a model so that given a new input X we try to find out the corresponding value of Y
- This forms the basis of supervised learning because for every test instance of X , we predict what must be the value of Y

Unsupervised Learning: scheme

DATA



CLUSTERS



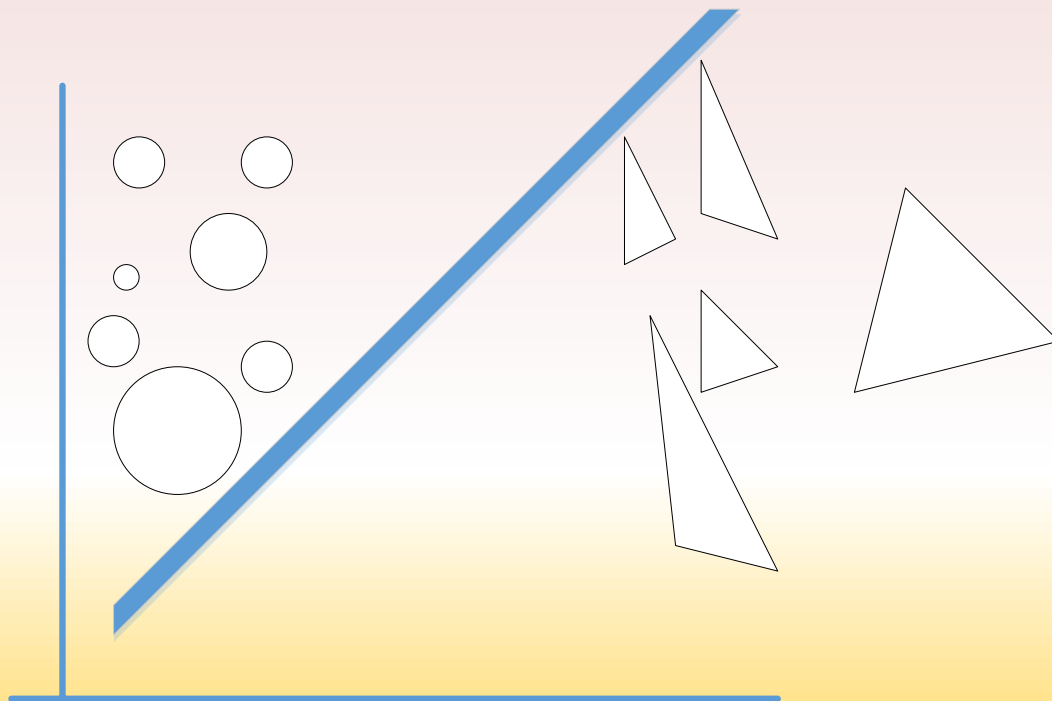
Unsupervised Learning: scheme

- In this learning we are provided with input X as data instances as the only data
- There is no label to the data
- The learning algorithm will produce clusters which group the data *based on the similarity between them* (depending on the features selected)
- Clusters summarize the data for find patterns in them

Semi-supervised Learning

- Here we have a combination of labelled and unlabelled data
- The labelled data is represented as classes of circles and triangles

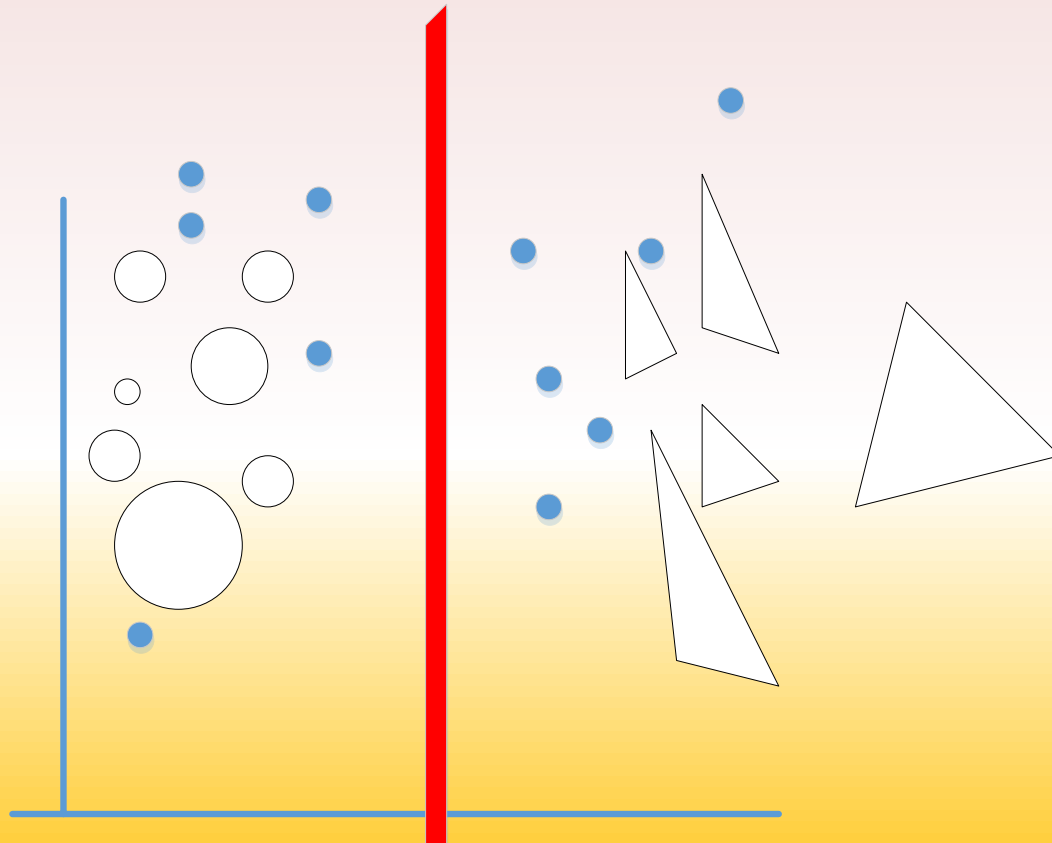
Semi-supervised Learning



Semi-supervised Learning

- It is a combination of supervised and unsupervised learning
- Labelled data along with larger amount of unlabelled training data
- This may be declared a solution with limitation on training data in supervised learning mode

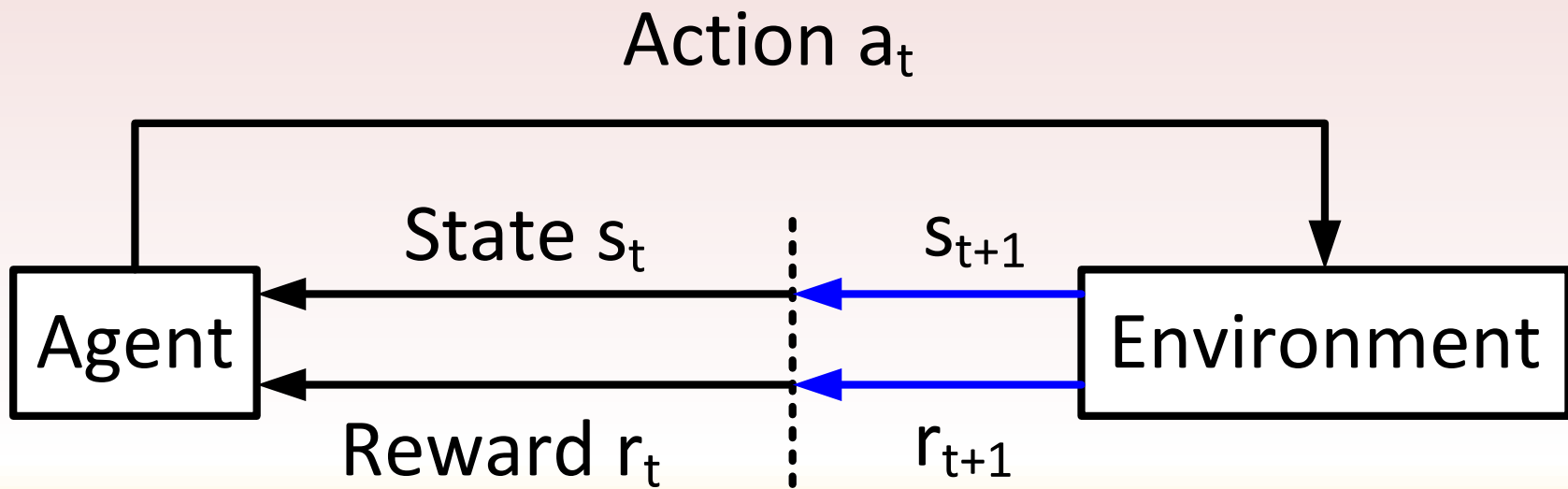
Semi-supervised Learning



Reinforcement Learning

- We have an agent that acts in the environment so that the agent takes an action and it can *impact* the environment
- So the agent is in a particular state, the agent takes an action and the *environment goes to a new state*. He gives some reward to the agent in new state
- The reward may be positive or negative (a penalty) and it can be nothing at that particular time step
- The agent is continually acting in this world

Reinforcement Learning: scheme



Resources: Datasets

- UCI Repository:

<http://www.ics.uci.edu/~mlearn/MLRepository.html>

- UCI KDD Archive:

<http://kdd.ics.uci.edu/summary.data.application.html>

- Statlib: <http://lib.stat.cmu.edu/>

- Delve: <http://www.cs.utoronto.ca/~delve/>

- Datasets:

https://www.researchgate.net/publication/391007555_Artificial_Intelligence_and_Machine_Learning_Datasets_Lab_09

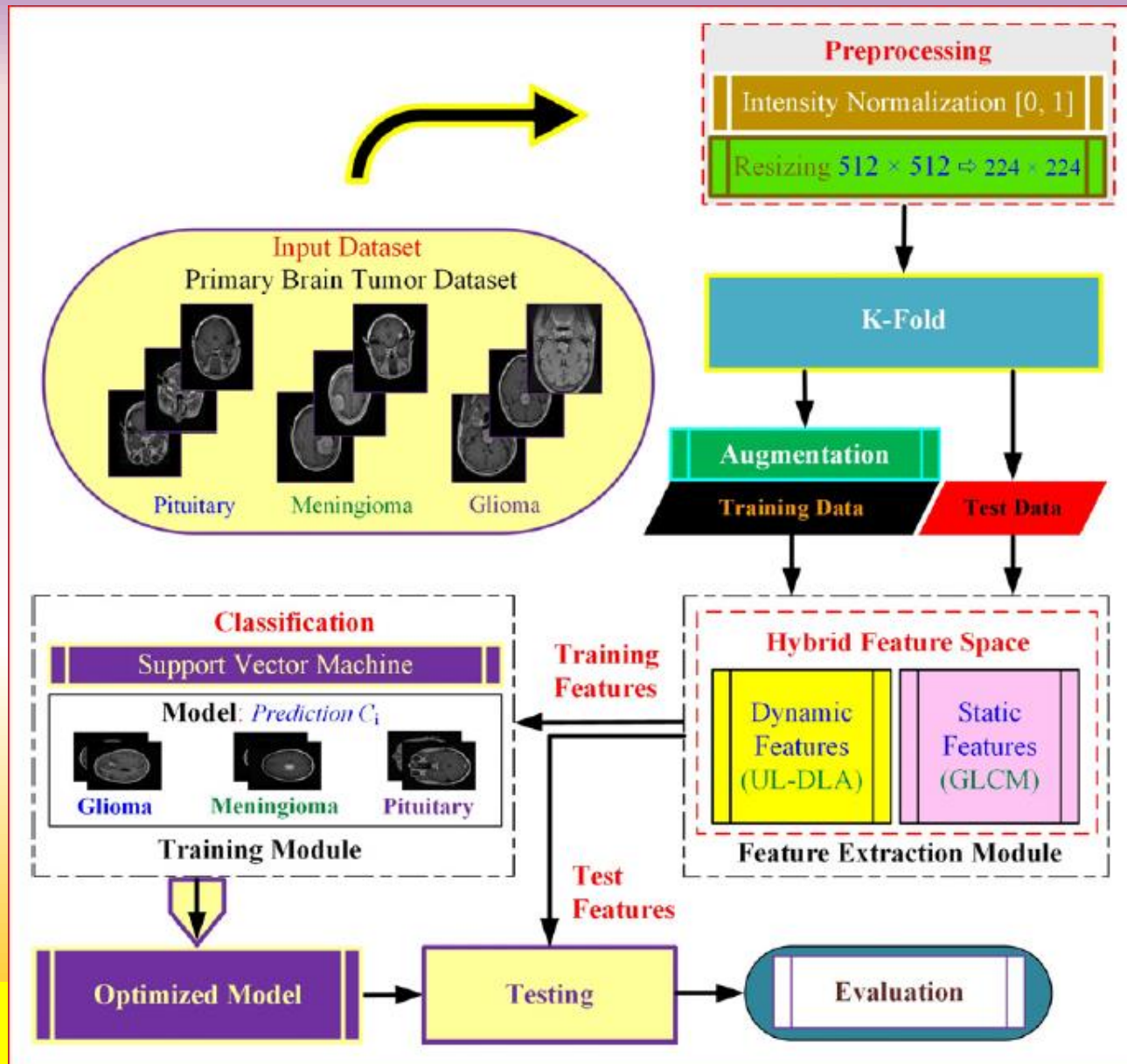
Challenges of ML

1. Data Quality and Availability
2. Bias and Fairness
3. Scalability
4. Generalization and Overfitting
5. Ethical Concerns

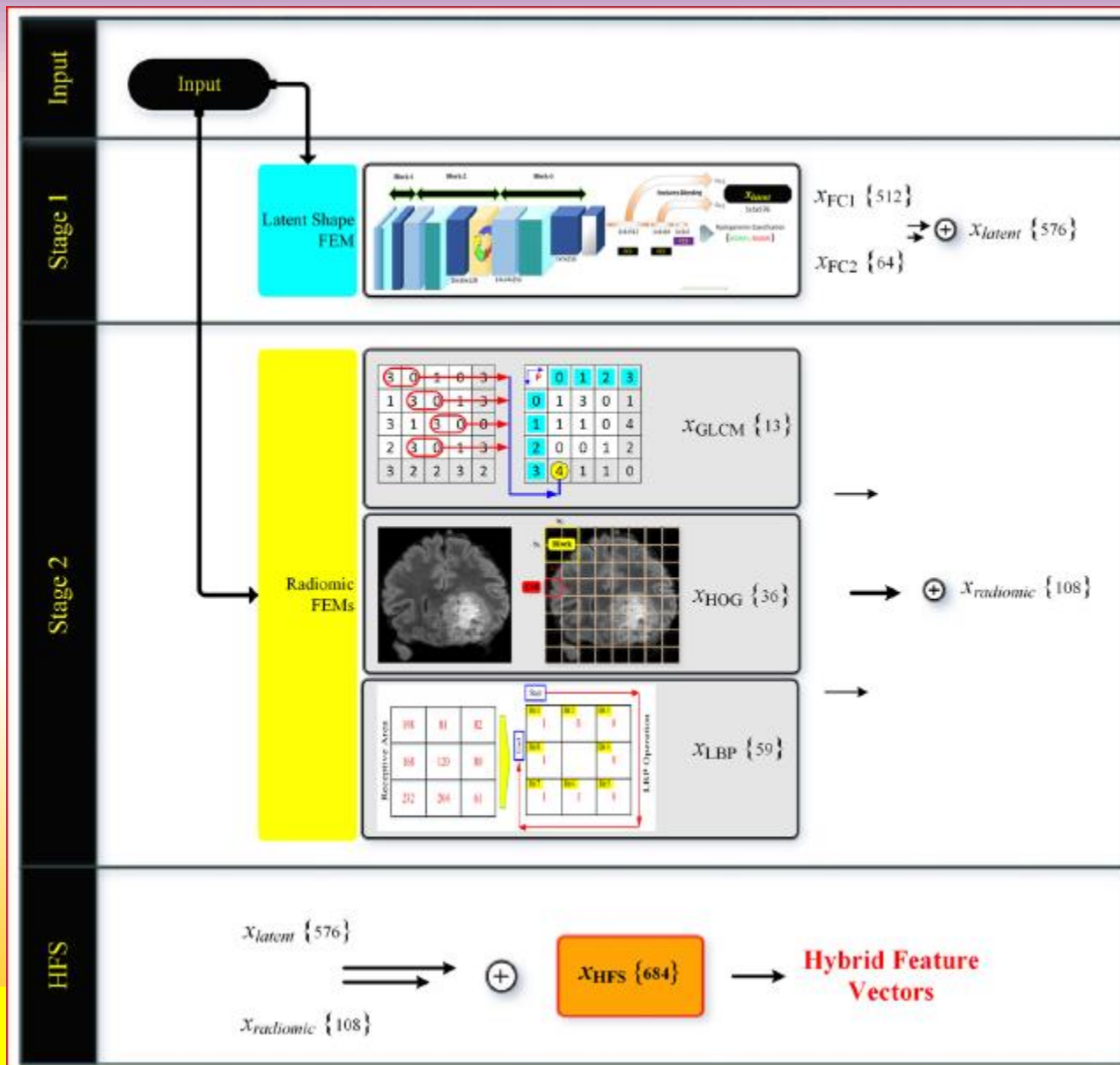
Future Trends of ML

1. Explainable AI (XAI)
2. Federated Learning
3. Edge AI and AI at the Edge
4. Ethical AI and Regulation
5. AI and Quantum Computing
6. AI-Powered Creativity
7. Transfer Learning and Pre-trained Models

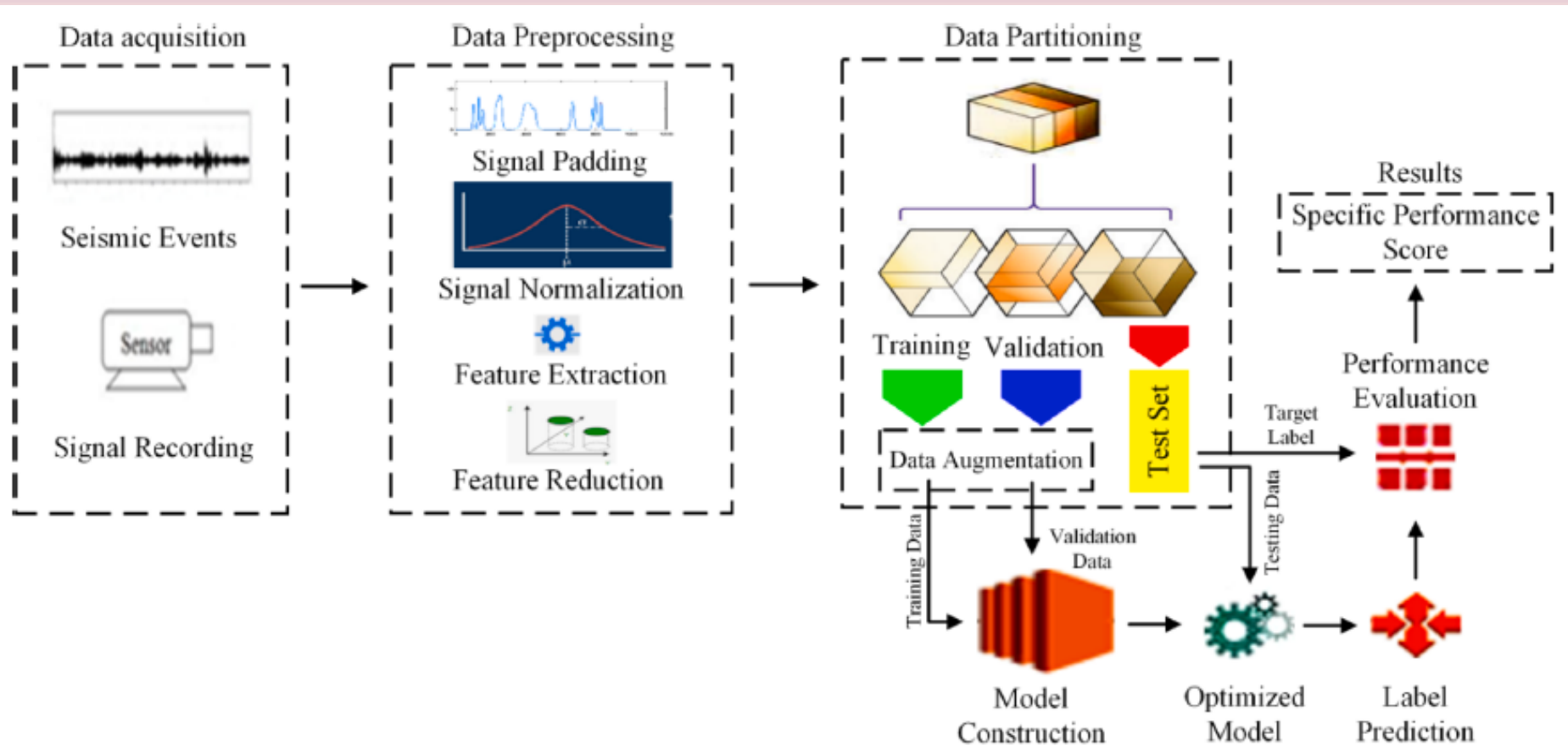
Modular Example 1



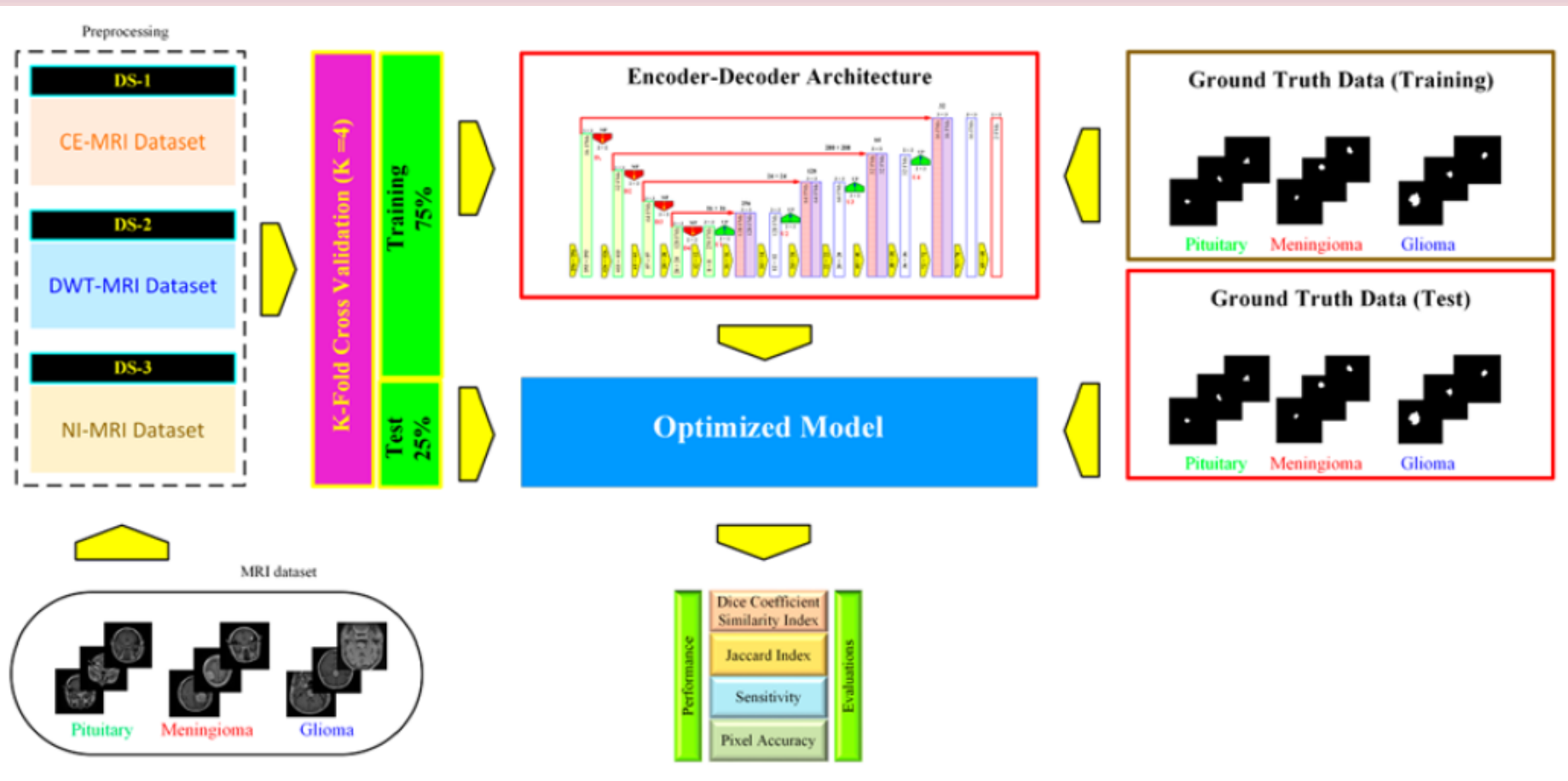
Modular Example 2



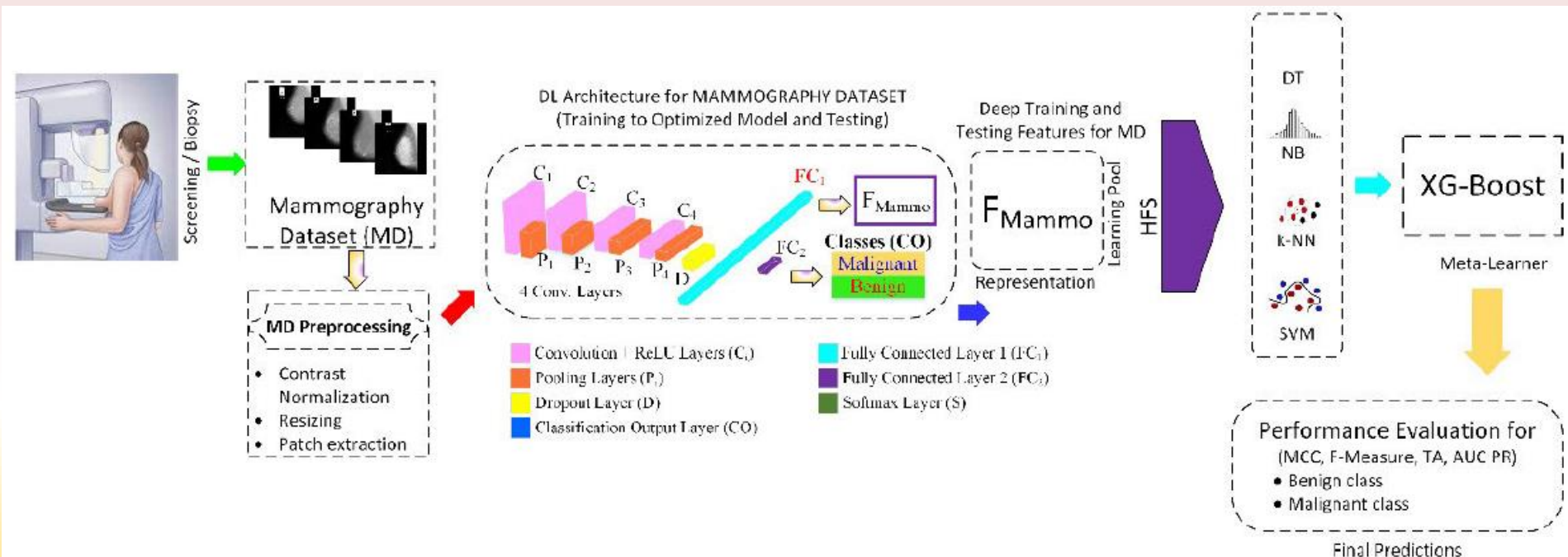
Modular Example 3



Modular Example 4



Modular Example 5





References



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